

Foundations of Arithmetic Power Geometry IV: The Coordinate-Dependent Defect Complex and the Modularity Barrier Obstruction Program

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Abstract

This paper establishes a rigorous analytical foundation for Arithmetic Power Geometry (APG) by formalising a scale-corrected geometric framework. By coupling the continuous path-integrated closure defect functional $A_0(a, b; p)$ to the logarithmic minimal discriminant $\ln |\Delta_E|$ of a semi-stable Frey curve $E_{a,b,p}$ associated with a hypothetical primitive Fermat triple, we construct the Scale-Corrected Global APG Invariant $\mathcal{A}_\Delta$. We prove an explicit global upper height bound mapping continuous deformation strain to absolute logarithmic Weil projective heights. Furthermore, we eliminate a major definitional vulnerability by proving the Local Coordinate-Dependent Damping Obstruction Theorem. We demonstrate that a universal rational damping constant K for the instantaneous defect field does not exist near the baseline. Instead, the local boundary condition is governed by a weight-dependent damping field that diverges to infinity under extreme coordinate skews. Finally, we formalise the modularity barrier roadmap, positioning the continuous analytic machinery as explicit structural motivation for a future obstruction program under the Ribet–Wiles regime.

Keywords: Arithmetic Power Geometry, Integrated Closure Defect Functional, Scale-Corrected Invariant, Shannon Entropy, Coordinate-Dependent Damping, Frey Curve, Faltings Height, Arithmetic Geometry, Modularity Barrier, Fermat’s Last Theorem.

1 Introduction and Literature Review

The historic proof of the modularity conjecture for semi-stable elliptic curves by Wiles (1995) and Taylor and Wiles (1995) established a profound connection between Diophantine equations and modular forms. This structural paradigm relies on the interplay between Galois representations and the arithmetic geometry of modular curves (Darmon et al., 1997; Diamond, 1996). The mechanism transitions through the properties of a hypothetical elliptic curve proposed by Frey (1986) and Hellegouarch (1972). Ribet (1990) proved that the mod- p Galois representation cannot exist at the required lower level. This framework leverages the geometric boundaries of modular curves elucidated by Mazur (1977, 1978) and Serre (1987).

Concurrently, the development of Diophantine geometry has been governed by heights and invariants (Faltings, 1983, 1984; Silverman, 1986). Arakelov (1974) and Szpiro (1981) introduced intersection theories on arithmetic surfaces to quantify the relationship between dis-

criminants and conductors. This framework expanded via bounds on rational points and height variations across Shimura varieties (Bombieri & Gubler, 2006; Hindry & Silverman, 2000; Lang, 1983; Vojta, 1987).

Arithmetic Power Geometry (APG), as introduced by Akhtar (2026a, 2026b, 2026c), introduces a dual perspective by treating exponents as continuous parameter deformations. While classical models fix exponents and vary fields, APG analyzes the continuous parameter landscape over the Power-Deformation Space $P_{(a,b)}$ centered at the Euclidean baseline $n = 2$ (Akhtar, 2026a). Akhtar (2026c) globalized this framework by introducing the scale-invariant path integral known as the Integrated Closure Defect Functional A_0 .

This paper addresses the scale blindspot of pure continuous deformation integrals by defining a combined product field. We prove the Local Coordinate-Dependent Damping Obstruction Theorem, demonstrating that a universal rational damping parameter is mathematically invalid because the local damping profile diverges under extreme skewness. This creates a rigorous analytic foundation to propose a formal pathway showing how continuous parameter deformations might govern the self-intersection mechanics of canonical sheaves, offering a potential link to the modular level drop limits verified by Abbes and Ullmo (1997), Saito (1988), and Zhang (1993).

2 Theoretical Foundations and the Scale-Corrected Invariant

Let $X_0(N)/\mathbb{Q}$ be the classical modular curve of level N . It is viewed as a suitable regular arithmetic model of $X_0(N)$ over $\text{Spec } \mathbb{Z}$ following Deligne (1971) and Shimura (1971). Let $J_1^t(\mathcal{X}_0(N))$ denote the first-order arithmetic jet bundle over the continuous exponent deformation parameter $t \in [2, p]$. Following the geometric principles of Katz (1973), we formalize the instantaneous closure defect field $D(t)$ from APG III (Akhtar, 2026c):

$$D(t) = 1 - \frac{a^t + b^t}{(a^2 + b^2)^{t/2}}.$$

The path integration of this field defines the scale-invariant Integrated Closure Defect Functional A_0 :

$$A_0(a, b; p) := \int_2^p D(t) dt.$$

To bridge continuous parameter dynamics and discrete number-theoretic magnitude, we introduce a unified structural invariant.

Definition 1 (Scale-Corrected Global APG Invariant). *Let $(a, b, c) \in (\mathbb{Z}^+)^3$ be a primitive, coprime integer triple satisfying a hypothetical Fermat relation at an odd prime exponent $p > 2$. Let*

$$E_{a,b,p} : y^2 = x(x - a^p)(x + b^p)$$

be the associated semi-stable Frey curve. The Scale-Corrected Global APG Invariant $\mathcal{A}_\Delta$ is defined as

$$\mathcal{A}_\Delta(a, b, c; p) := A_0(a, b; p) \cdot \ln |\Delta_E|.$$

3 Proved Analytical Theorems and Bounds

Theorem 1 (Baseline Scale-Corrected Upper Height Bound). *Let $(a, b, c) \in (\mathbb{Z}^+)^3$ be a primitive, coprime triple of positive integers. For any odd prime exponent $p > 2$, the scale-corrected global APG invariant satisfies the strict upper boundary*

$$\mathcal{A}_\Delta(a, b, c; p) < 6p(p-2) \cdot h(a : b : c),$$

where

$$h(a : b : c) = \ln \max(a, b, c)$$

is the absolute logarithmic Weil projective height.

Proof. By the strict monotonicity of ℓ_p -norms on $(0, \infty)$ for non-zero coordinate vectors, we have

$$(a^t + b^t)^{1/t} < (a^2 + b^2)^{1/2}$$

for all $t > 2$. Raising this relation to the power of t preserves the strict structural inequality:

$$\frac{a^t + b^t}{(a^2 + b^2)^{t/2}} < 1 \implies 0 < D(t) < 1 \quad \forall t \in (2, p].$$

We evaluate the standard Riemann integral of this bounded field across the compact integration domain $[2, p]$:

$$A_0(a, b; p) = \int_2^p D(t) dt < \int_2^p 1 dt = p - 2.$$

The minimal discriminant Δ_E of the semi-stable Frey curve associated with a hypothetical primitive Fermat triple scales as

$$|\Delta_E| \leq (abc)^{2p}.$$

Because

$$abc < [\max(a, b, c)]^3,$$

taking the natural logarithm translates this into absolute height space:

$$\ln |\Delta_E| \leq 2p \ln(abc) < 6p \ln \max(a, b, c) = 6p \cdot h(a : b : c).$$

Multiplying these independent, strictly positive upper bounds yields the compound upper ceiling:

$$\mathcal{A}_\Delta(a, b, c; p) = A_0(a, b; p) \cdot \ln |\Delta_E| < 6p(p-2) \cdot h(a : b : c).$$

□

Theorem 2 (Local Coordinate-Dependent Damping Obstruction Theorem). *Let $(a, b) \in (\mathbb{Z}^+)^2$ be a coprime coordinate pair with normalized baseline weights*

$$w_a = \frac{a^2}{a^2 + b^2} \quad \text{and} \quad w_b = \frac{b^2}{a^2 + b^2}.$$

Let

$$H(W) = -(w_a \ln w_a + w_b \ln w_b)$$

represent the Shannon coordinate entropy, and let

$$Q(a, b) = w_a (\ln w_a)^2 + w_b (\ln w_b)^2$$

represent the quadratic logarithmic weight dispersion parameter. Then, a universal constant $K > 0$ satisfying the bound

$$D(t) \geq \frac{\frac{H(W)}{2}(t-2)}{1+K(t-2)} \quad \forall t \geq 2$$

does not exist near the baseline $t \rightarrow 2^+$. Instead, near the baseline, the parameter framework obeys a tight, structurally dynamic rational boundary where the minimal damping factor is regulated by coordinate skewness:

$$K(W) \geq \frac{Q(a, b)}{4H(W)}.$$

Furthermore, as the coordinate support becomes infinitely skewed ($w_a \rightarrow 0$), the local damping parameter expands as

$$-\frac{1}{4} \ln w_a,$$

causing it to diverge to infinity.

Proof. The continuous closure defect functional integrand can be written in exponential base form over normalized coordinate weights:

$$D(t) = 1 - \left(w_a^{t/2} + w_b^{t/2} \right).$$

Let

$$x = \frac{t-2}{2},$$

which implies

$$\frac{t}{2} = 1 + x.$$

As the parameter path nears the Pythagorean baseline ($t \rightarrow 2^+$), $x \rightarrow 0^+$. We rewrite the instantaneous defect field:

$$D(t) = 1 - (w_a^{1+x} + w_b^{1+x}) = 1 - w_a e^{x \ln w_a} - w_b e^{x \ln w_b}.$$

Applying a standard Maclaurin series expansion to the exponential components,

$$e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + O(u^4),$$

yields

$$w_a e^{x \ln w_a} = w_a \left[1 + x \ln w_a + \frac{x^2}{2} (\ln w_a)^2 + O(x^3) \right],$$

and

$$w_b e^{x \ln w_b} = w_b \left[1 + x \ln w_b + \frac{x^2}{2} (\ln w_b)^2 + O(x^3) \right].$$

Summing these components and applying the total mass conservation constraint $w_a + w_b = 1$ isolates the higher-order parameters:

$$\begin{aligned} D(t) &= 1 - (w_a + w_b) - x(w_a \ln w_a + w_b \ln w_b) \\ &\quad - \frac{x^2}{2} (w_a (\ln w_a)^2 + w_b (\ln w_b)^2) + O(x^3). \end{aligned}$$

Thus

$$D(t) = xH(W) - \frac{x^2}{2}Q(a, b) + O(x^3).$$

Substituting $x = (t-2)/2$ back into the equation yields the exact second-order Taylor series representation near the Euclidean baseline:

$$D(t) = \frac{H(W)}{2}(t-2) - \frac{Q(a, b)}{8}(t-2)^2 + O((t-2)^3).$$

We require the instantaneous defect field to be bounded below by a rational function. We set up the inequality to solve for the boundary threshold condition of K near the baseline:

$$D(t) \geq \frac{\frac{H(W)}{2}(t-2)}{1 + K(t-2)}.$$

Hence

$$\frac{1}{D(t)} \leq \frac{1 + K(t-2)}{\frac{H(W)}{2}(t-2)} = \frac{2}{H(W)(t-2)} + \frac{2K}{H(W)}.$$

Rearranging the terms isolates the required parameter condition for K :

$$2K \geq \frac{H(W)}{D(t)} - \frac{2}{t-2},$$

or equivalently,

$$K \geq \frac{H(W)}{2D(t)} - \frac{1}{t-2}.$$

We evaluate the local behavior of this required limit as the trajectory approaches the baseline ($t \rightarrow 2^+$) by substituting our derived Taylor series representation:

$$K \geq \lim_{t \rightarrow 2^+} \left[\frac{H(W)}{2 \left(\frac{H(W)}{2}(t-2) - \frac{Q(a, b)}{8}(t-2)^2 + \dots \right)} - \frac{1}{t-2} \right].$$

Factoring out the linear component $(t-2)$ from the denominator gives

$$K \geq \lim_{t \rightarrow 2^+} \left[\frac{1}{(t-2) \left(1 - \frac{Q(a, b)}{4H(W)}(t-2) + \dots \right)} - \frac{1}{t-2} \right].$$

Applying a geometric series expansion,

$$\frac{1}{1-u} = 1 + u + O(u^2),$$

to the interior bracket yields

$$K \geq \lim_{t \rightarrow 2^+} \left[\frac{1}{t-2} \left(1 + \frac{Q(a,b)}{4H(W)}(t-2) + \dots \right) - \frac{1}{t-2} \right] = \frac{Q(a,b)}{4H(W)}.$$

To analyze the global properties under extreme coordinate skewness, we take the asymptotic limit as $w_a \rightarrow 0^+$, which forces $w_b \rightarrow 1^-$. The constituent parameters collapse as

$$H(W) \sim -w_a \ln w_a$$

and

$$Q(a,b) \sim w_a (\ln w_a)^2.$$

Substituting these dominant tracking structures into our proved threshold inequality yields

$$K(W) \geq \frac{w_a (\ln w_a)^2}{4(-w_a \ln w_a)} = -\frac{1}{4} \ln w_a.$$

As $w_a \rightarrow 0^+$, the term

$$-\frac{1}{4} \ln w_a \rightarrow \infty.$$

Because the local damping factor grows arbitrarily large, no single universal constant K can protect the rational lower envelope near the baseline across the entire coordinate space. $\square$

4 The Modularity Barrier and Conjectural Obstruction Program

To transition this validated analytical core into an arithmetic tool, the proved bounds must be linked directly to the ramification behavior of modular Galois representations. We formalize the remaining intermediate milestones required to bridge this barrier.

Conjecture 1 (The Arakelov Spectral Jet Lower Envelope). *Let $\mathbb{G}_{\text{jet}}(z, s; t)$ be the smooth solution on $X_0(N)(\mathbb{C}) \times \mathbb{R}^+$ to the perturbed Laplace–Beltrami equation*

$$[\Delta_{X_0(N)} - t(t-1)]\mathbb{G}_{\text{jet}}(z, s; t) = \delta_z(s) \cdot D(t),$$

whose path integral defines the global jet-strain tensor

$$\Xi_{\text{APG}}(z) = \int_2^p \mathbb{G}_{\text{jet}}(z, z; t) dt.$$

Then, Ξ_{APG} satisfies a strict lower bound governed by the weight-dependent damping parameter $K(W)$:

$$\Xi_{\text{APG}}(z) \geq \frac{H(W) \cdot (p-2)^2}{4 \cdot [1 + K(W)(p-2)]} \cdot \text{vol}(X_0(N)).$$

Analytical Strategy. By the spectral decomposition of the hyperbolic Laplacian on $X_0(N)(\mathbb{C})$ established by Gross and Zagier (1986), the kernel function expands via Maass wave forms ϕ_j . Isolating the constant leading eigenfunction ϕ_0 yields

$$|\phi_0(z)|^2 = \frac{1}{\text{vol}(X_0(N))}$$

with $\lambda_0 = 0$. Proving Conjecture 1 requires showing that the variable field

$$K(W) \geq \frac{Q(a, b)}{4H(W)}$$

derived in Theorem 2 serves as a uniform lower bound for the remaining positive non-vanishing spectrum.

Conjecture 2 (The Arithmetic Intersection and Faltings Height Bridge). *The global spectral jet-strain tensor Ξ_{APG} forces a strict arithmetic lower bound on the stable Faltings height of the Frey curve associated with a hypothetical primitive Fermat triple:*

$$h_F(E_{a,b;p}) \geq \frac{1}{12} \int_{X_0(N)} \Xi_{\text{APG}}(z) d\mu_{\text{Hyp}}(z).$$

Geometric Strategy. Faltings (1983, 1984) links the Faltings height to the arithmetic self-intersection number

$$\widehat{c}_1(\overline{\omega}_{\mathcal{X}_0(N)/\mathbb{Z}})^2.$$

This program requires demonstrating that deforming the hyperbolic metric via the conformal factor

$$e^{2\Xi_{\text{APG}}(z)}$$

forces an expansion of the self-intersection of the canonical sheaf across the regular minimal model criteria (Saito, 1988).

Conjecture 3 (The APG Mod- p Modularity Program). *Under the continuous parameter trajectory $t \in [2, p]$, the attached mod- p Galois representation*

$$\rho_{E,p}$$

cannot satisfy the conditions required to undergo modular level-lowering down to level $N' = 2$.

Motivation for a Future Obstruction Theorem. Ken Ribet's (1990) Level-Lowering Theorem enforces strict ramification conditions on the modular Galois representation. If the height can be bounded from above by a conjectural modular curve ceiling, intersecting the lower bounds from Conjecture 3 and Conjecture 4 against this ceiling would yield

$$\frac{H(W) \cdot (p-2)^2}{48 \cdot [1 + K(W)(p-2)]} \leq \frac{\ln(2) + O(\ln p)}{[\text{rad}(abc)]^2}.$$

Because the left-hand side grows linearly while the right-hand side collapses exponentially

due to discriminant growth

$$|\Delta_E| \asymp (abc)^{2p},$$

completing proofs for Conjectures 3 and 4 would establish an asymptotic divergence. This behavior provides the necessary motivation for a future obstruction theorem to block the level drop and eliminate the hypothetical Frey curve for high exponents.

5 Conclusion and Strategic Status

APG IV introduces the scale-corrected global invariant $\mathcal{A}_\Delta$ and proves its baseline upper height ceiling. Furthermore, it establishes a genuine analytic milestone by proving Theorem 2, showing that a universal rational damping constant is mathematically invalid due to local divergence under extreme coordinate skewness.

However, the framework remains positioned as a rigorous scale-corrected APG IV roadmap and analytic framework, not as a completed proof of Fermat’s Last Theorem. It identifies, but does not yet solve, the modular ramification bridge required for an APG proof of FLT. The complete proof engine remains conditional on resolving the core mathematical transformations between continuous deformation strain and discrete Galois representations outlined in Conjectures 1–3.

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